

ADAPTIVE HYBRID TUPLE-PARTITIONING

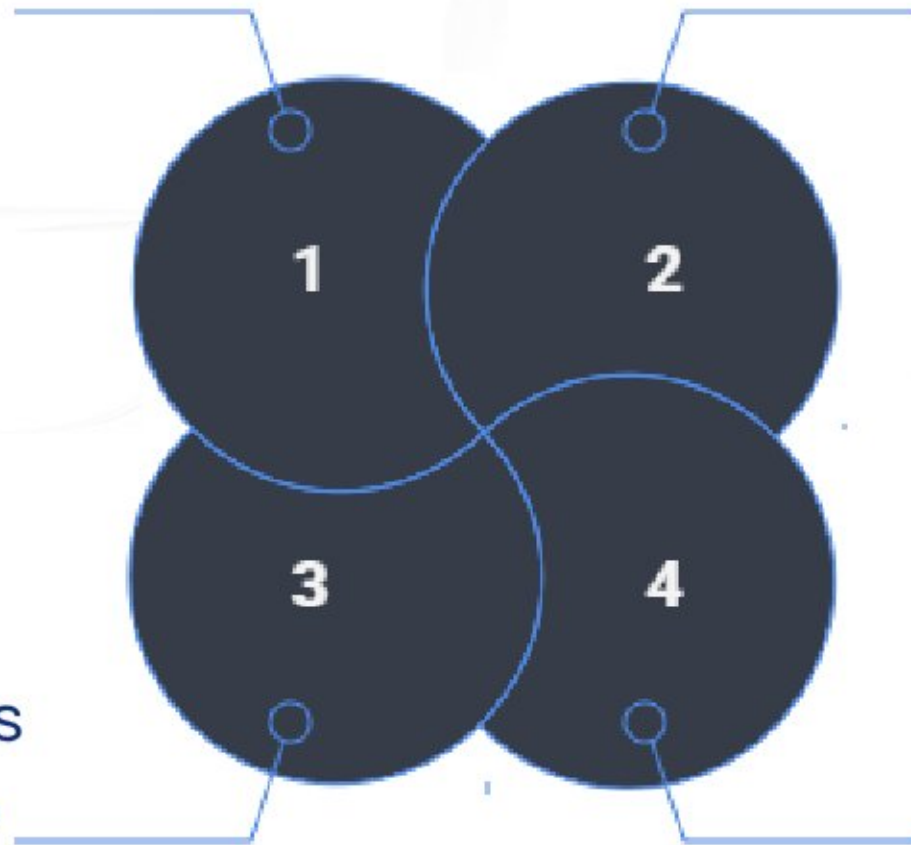
FOR LARGE DISTRIBUTED DBS



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PROBLEM & KEY CHALLENGES

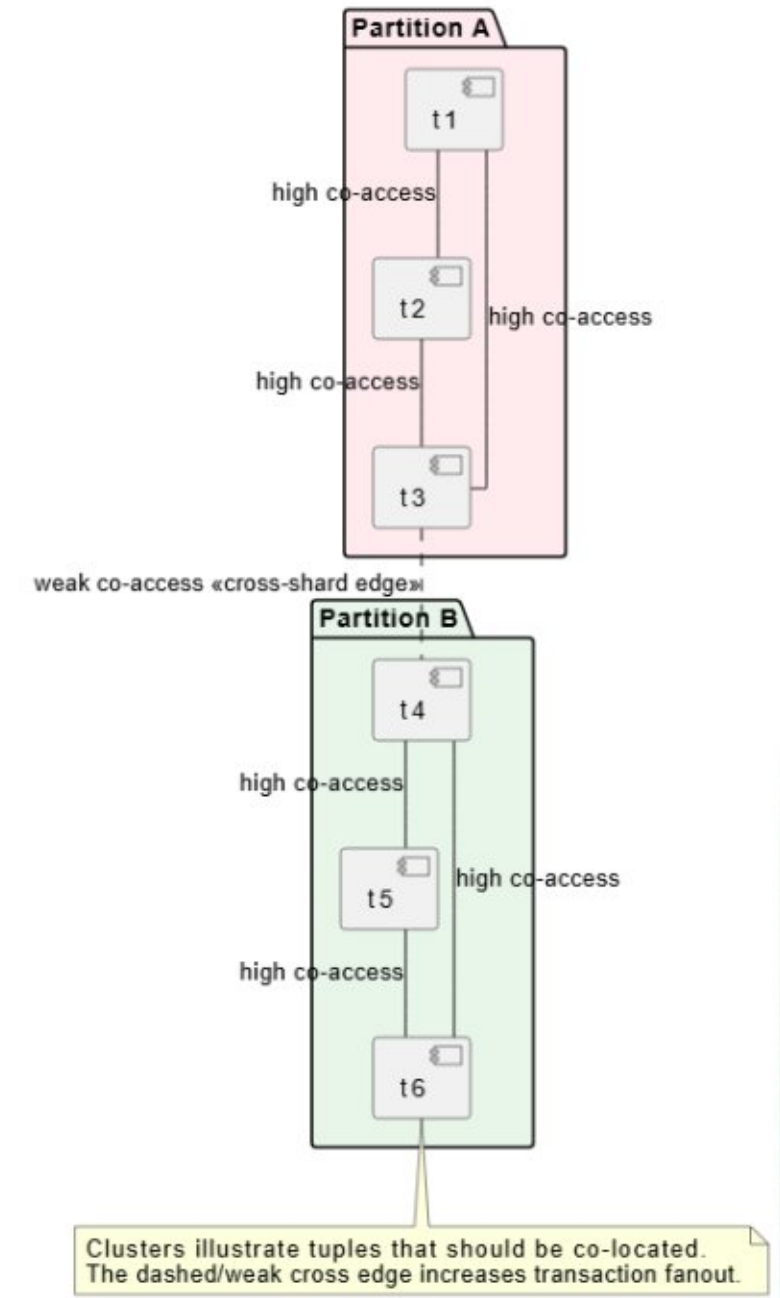
Goal:
reduce distributed
transactions while
maintaining
availability and
bounding migration
cost.



Operational constraints:
low runtime overhead,
bounded migrations,
simple deployable
stack.

OLTP scale limited by
distributed transactions
(network coordination,
cross-site locking, tail
latency).

Research challenge:
adapt quickly to
workload drift without
expensive global
recomputation.



HYBRID APPROACH COMPONENTS

Periodic learned refinement

GNN or regressor proposes local updates, accepting only safe ones.



Hot-tuple handling

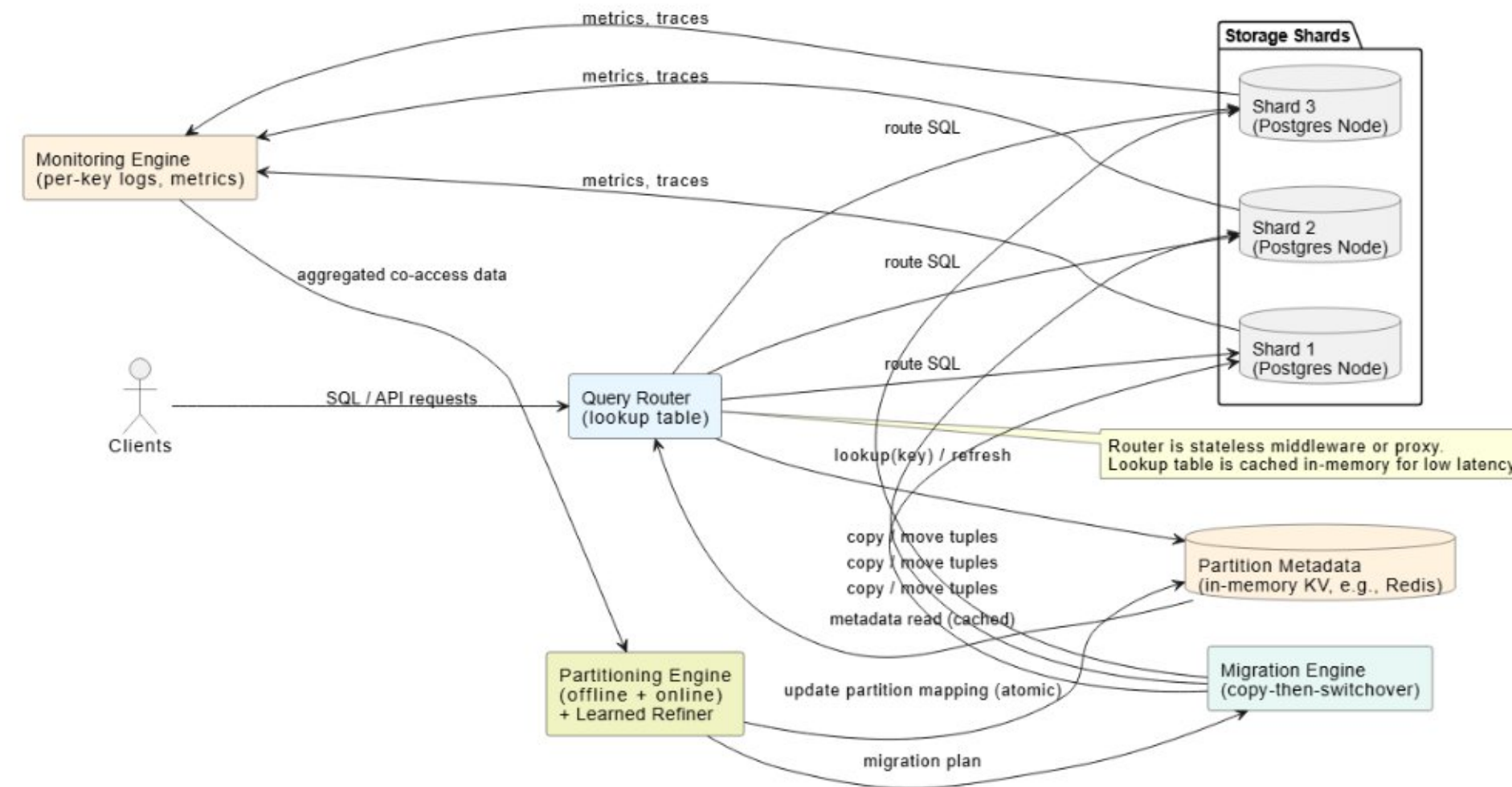
Copy-then-switchover, pinning, and write-forwarding techniques.

Offline baseline

Initial partition created using Schism or METIS.

Lookup router

In-memory key-to-shard mapping used by stateless proxy.



MATHEMATICAL OBJECTIVE

COST OBJECTIVE (FORMAL)

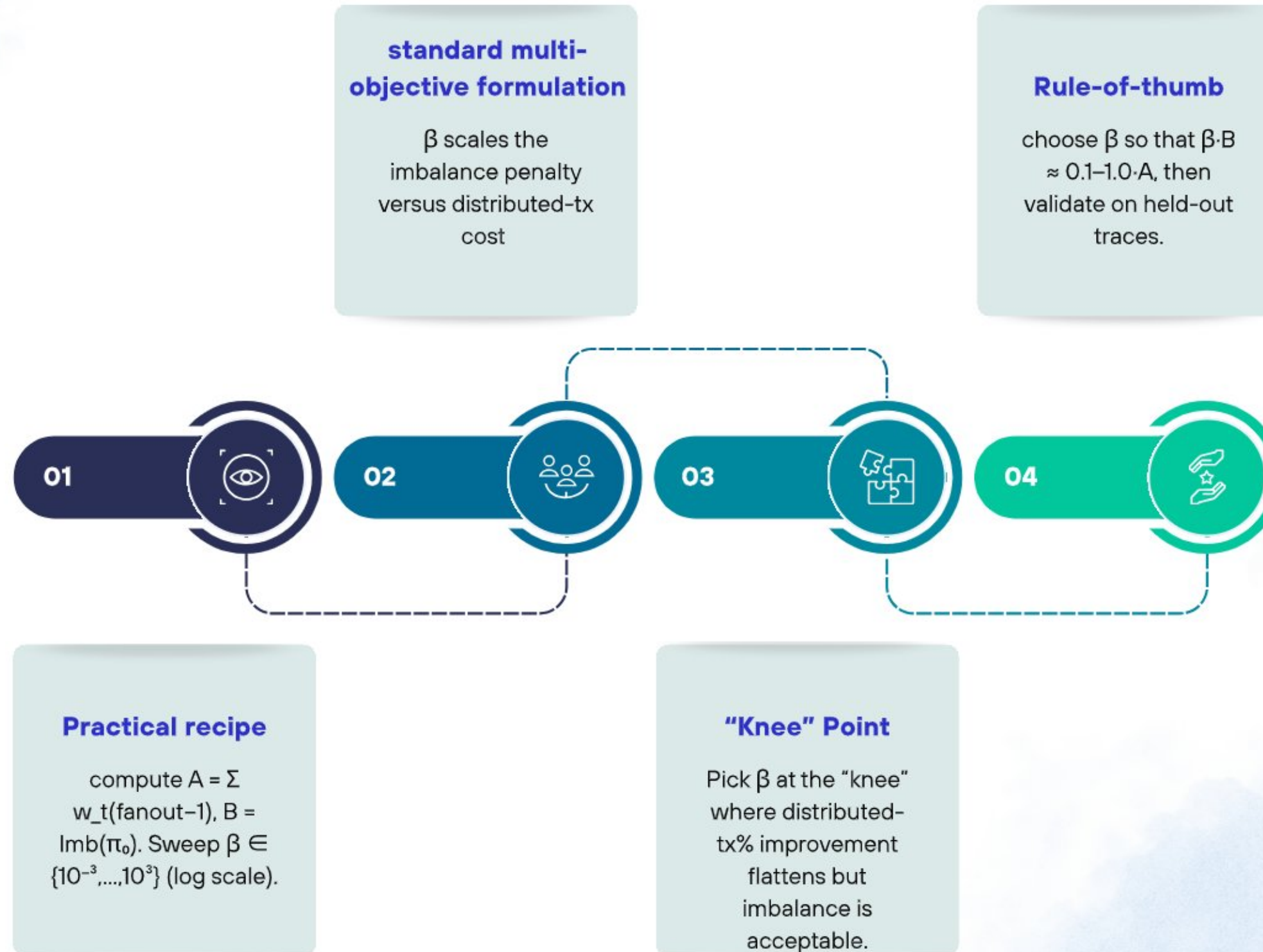
□ — $\text{fanout}(t, \pi) = |\{\pi(i) : i \in t\}|.$

□ — $\text{Cost}(\pi) = \sum_{t \in \mathcal{T}} w_t \cdot (\text{fanout}(t, \pi) - 1) + \beta \cdot \text{Imb}(\pi),$

□ — $\text{Imb}(\pi) = \max_p L_p - \frac{1}{K} \sum_v L_p,$

- First term: aggregated penalty for multi-shard transactions (frequency weighted).
- β : tradeoff scalar controlling emphasis on load balance vs. distributed-tx reduction.

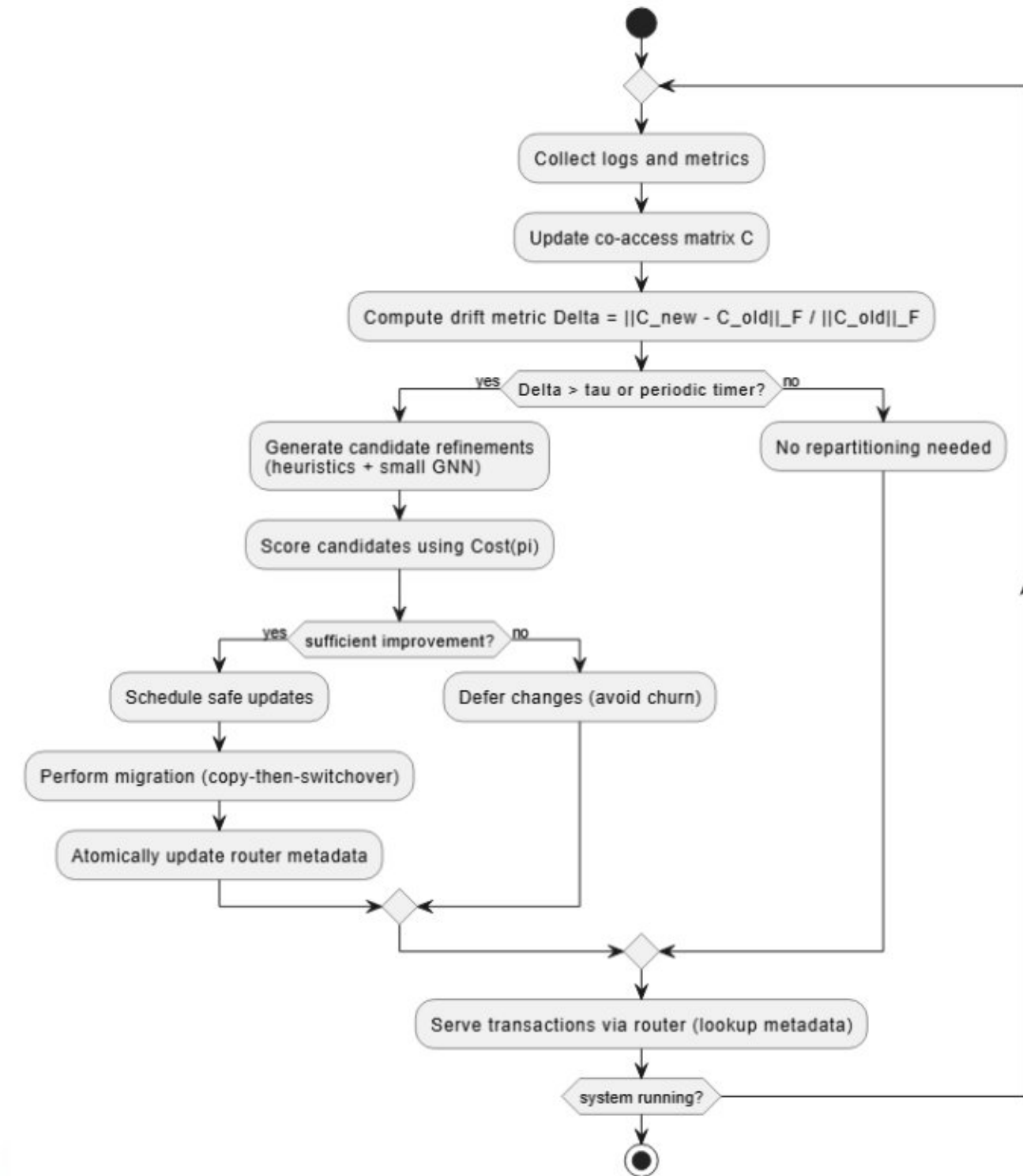
INTERPRETING & SELECTING β



DRIFT DETECTION & SLIDING WINDOW

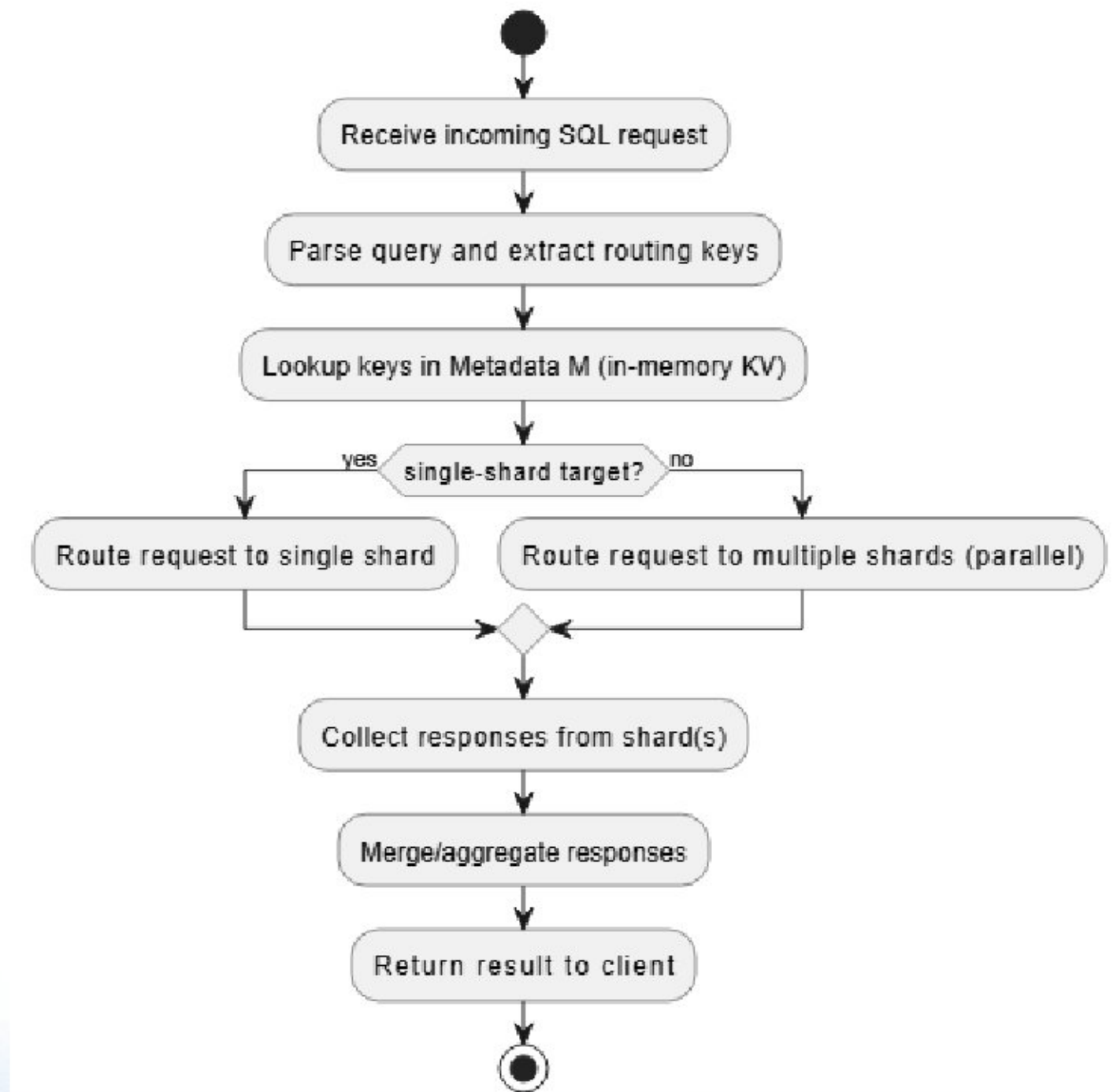
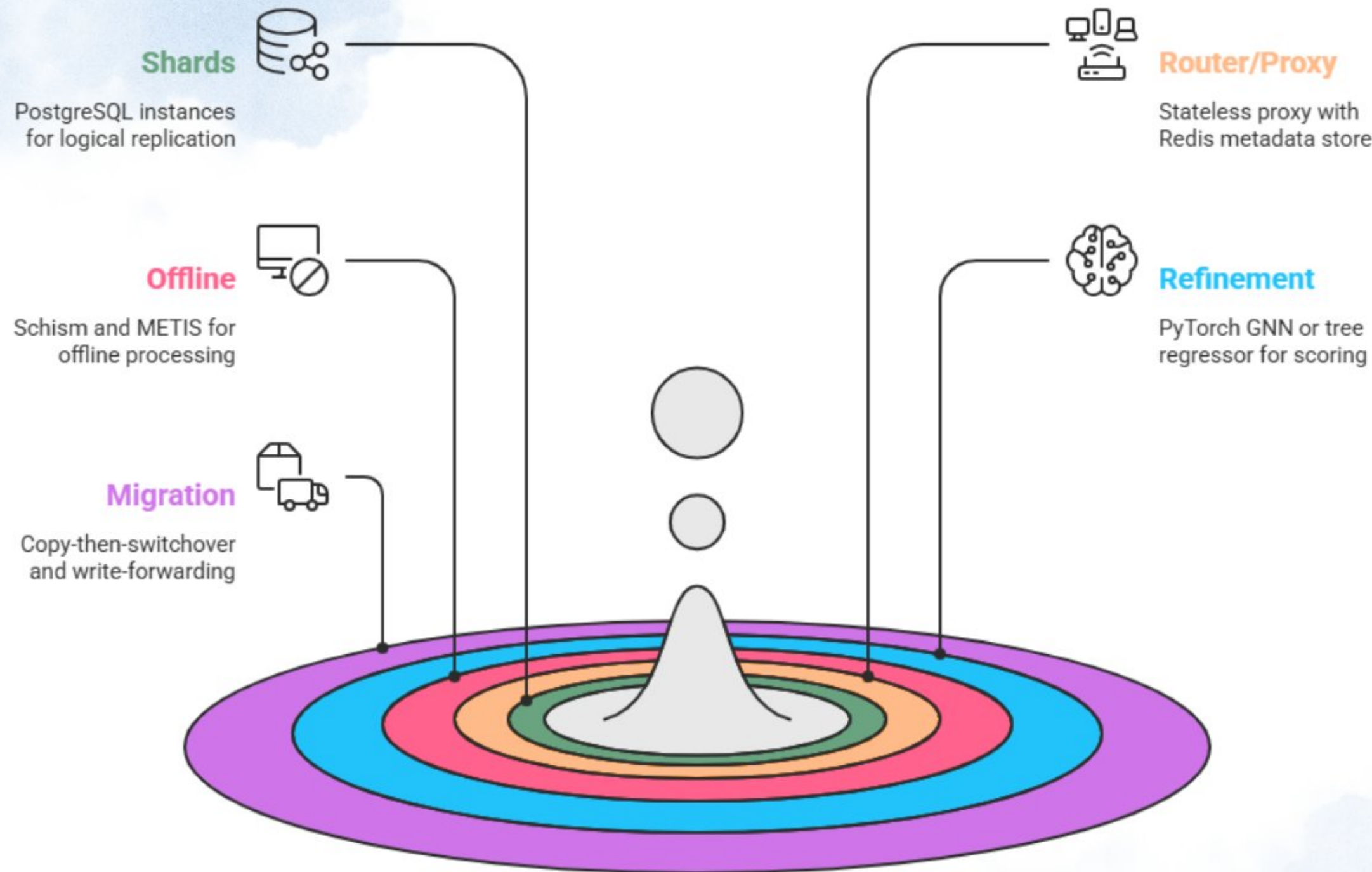
$$\Delta = \frac{\|C_{W_2} - C_{W_1}\|_F}{\|C_{W_1}\|_F},$$

- Monitor via lightweight sampling (1–5% of transactions) and per-key counters.
- Drift metric: $\Delta = \|C_{W_2} - C_{W_1}\|_F / \|C_{W_1}\|_F$. Trigger when $\Delta > \tau$.
- Window choices: 1 min for bursty; 5–15 min for stable workloads.
- Choose τ from baseline ($\tau = \mu_\Delta + 2\sigma_\Delta$) or fixed relative change (10–30%).
- Accept candidate updates only if Cost reduces by $> \varepsilon$ ($\varepsilon = 1$ –5%).



PROTOTYPE STACK

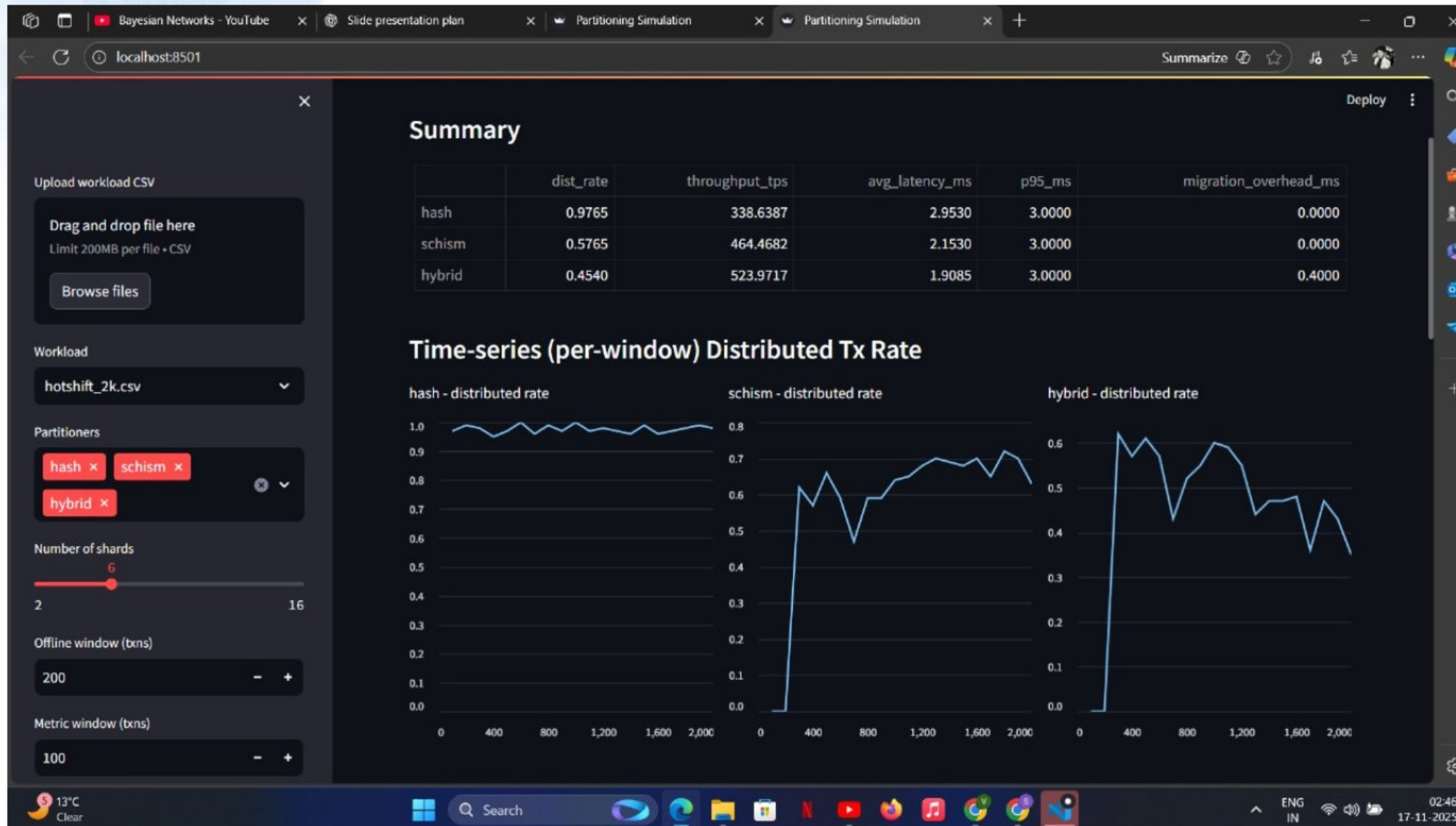
FEASIBILITY



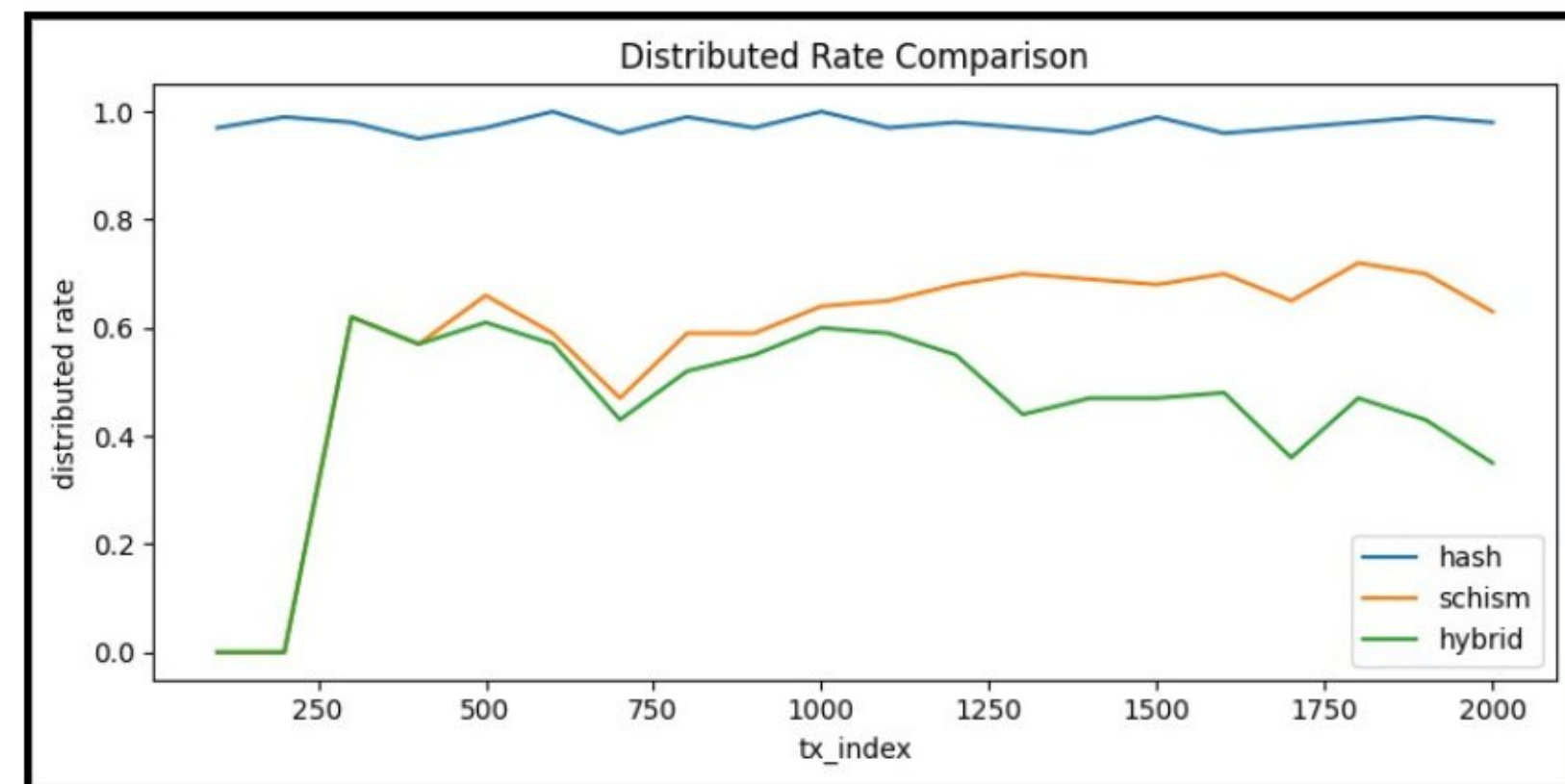
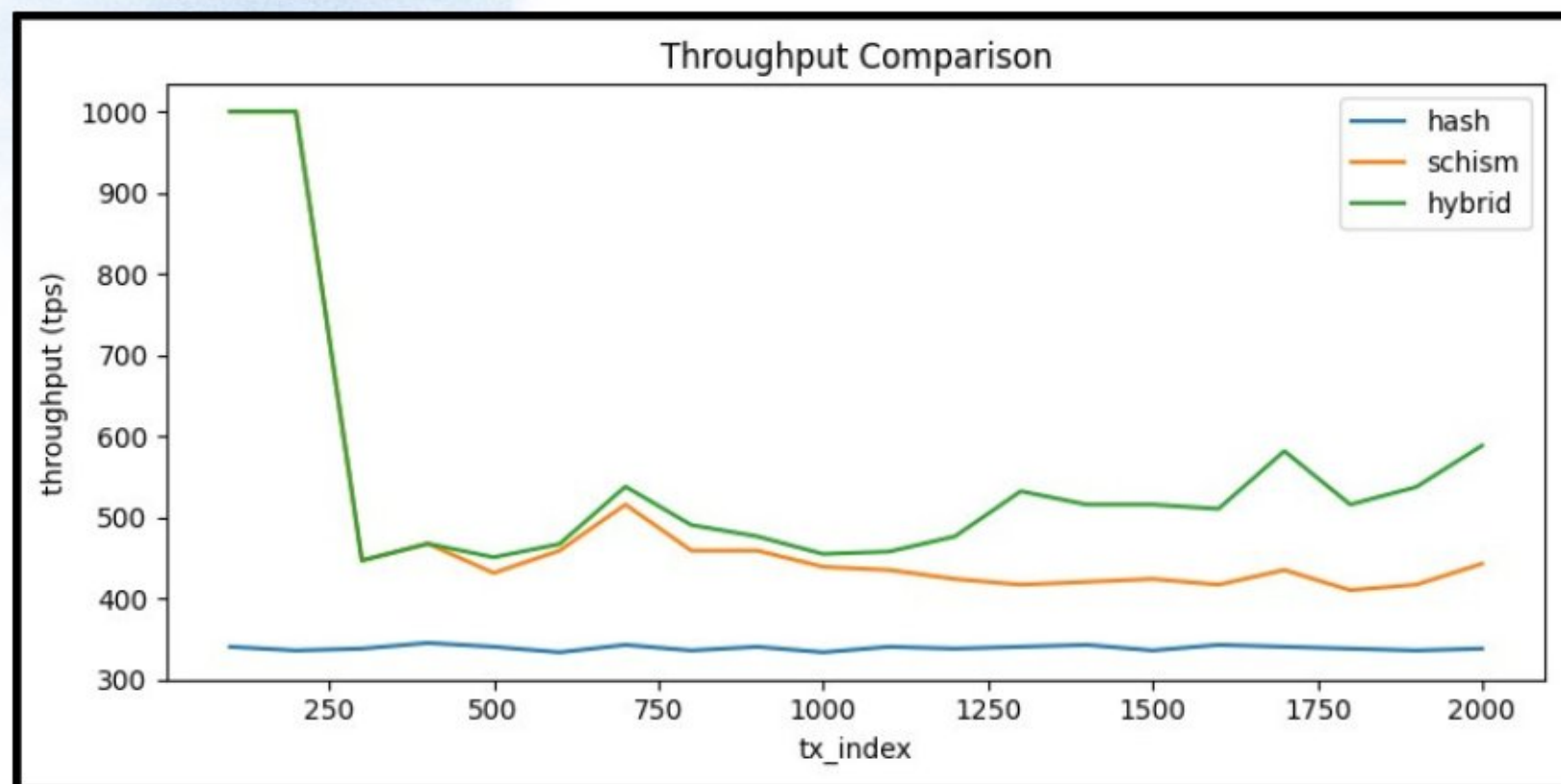
EVALUATION PLAN & METRICS

- **Workloads:** TPC-C like traces; hotspot-shift traces to test drift adaptation.
- **Metrics:** % distributed transactions, throughput (TPS), tail latencies (p95/p99), migration overhead (bandwidth, CPU), stability (#repartitions).
- **Baselines:** Hash, Schism (offline), Proposed Hybrid.

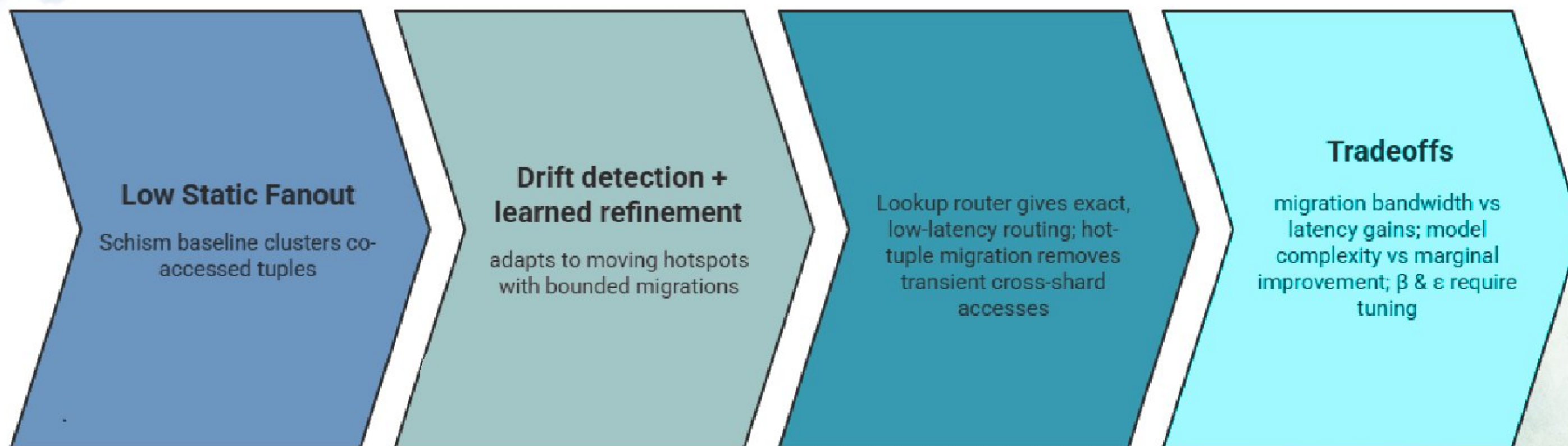
RESULTS



RESULTS



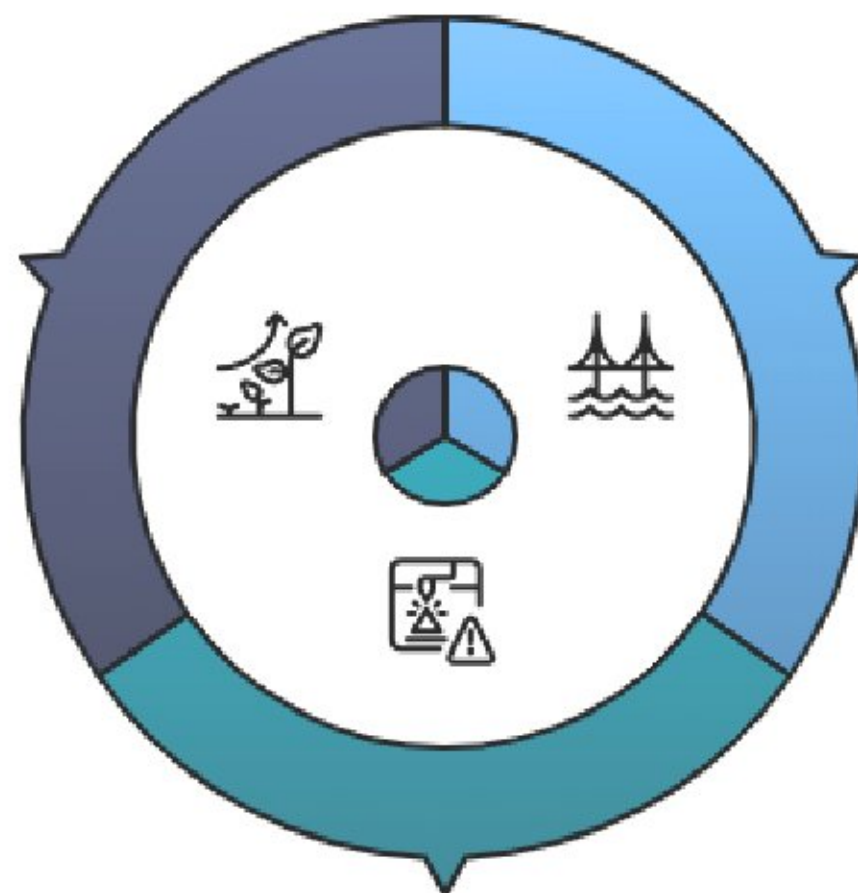
ANALYSIS & TRADEOFFS



NOVELTY, LIMITATIONS & NEXT STEPS

Next steps

Auto-tune β , latency-aware cost model, geo-aware partitioning.
Finalize sensitivity studies, publish plots.



Novelty

Safe updates and throttled migration for production. Bridges offline partitioning and online adaptation.

Limitations

Write-heavy workloads stressed by migration.
Model needs representative data, geo-replication not covered.



THANK YOU